Project Design Phase-I
Solution Architecture

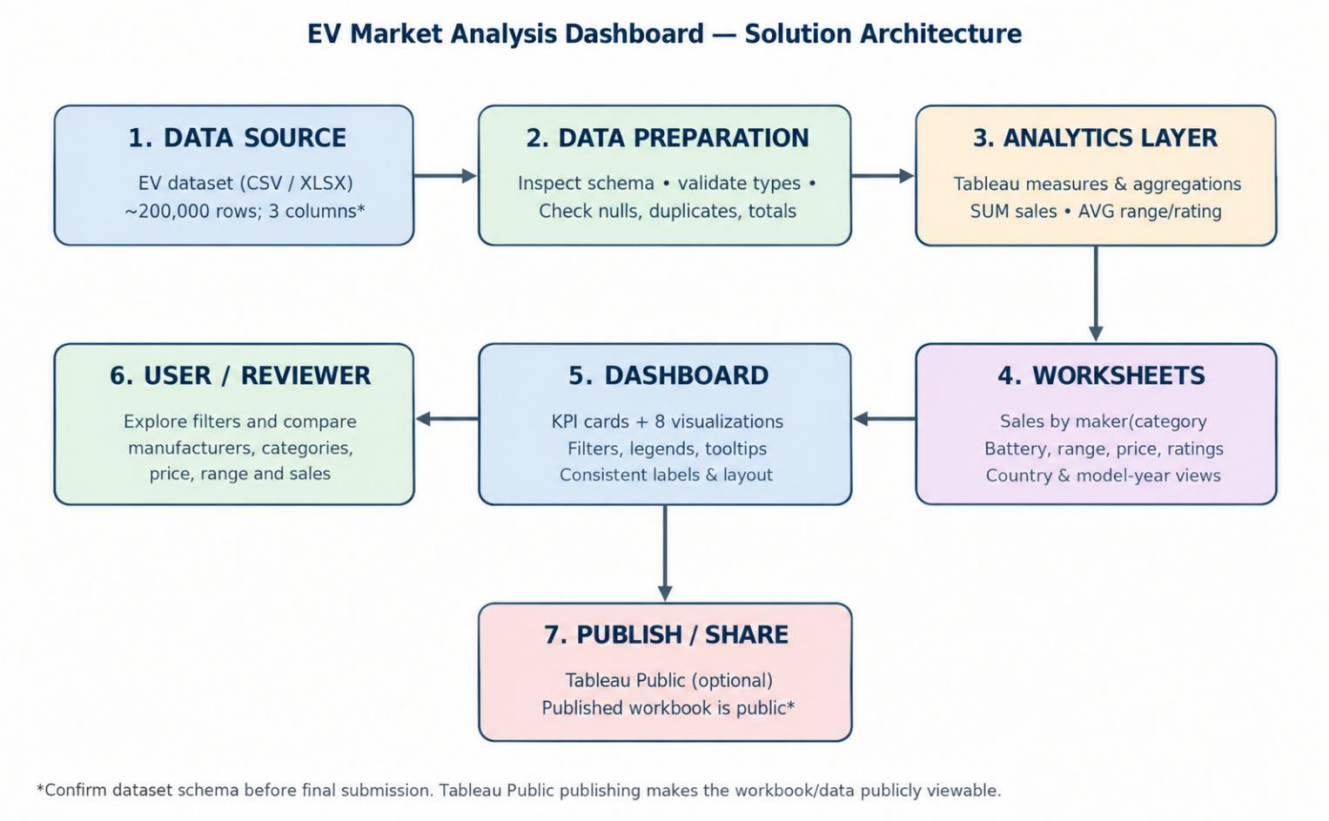
Date	16 October 2026
Team ID	Roll 58
Project Name	EV Market Project design
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behaviour, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram — EV Market Analysis Dashboard



Data flow: EV dataset → data validation/preparation → Tableau measures → individual worksheets → interactive dashboard → user exploration and optional Tableau Public sharing.

Component Name	Description/Role in Architecture	Technology Used
Data source	Stores input EV market records and attributes.	CSV/XLSX dataset
Data validation	Checks column names, data types, missing values, duplicates and unusual values.	Spreadsheet checks; optional pandas/NumPy
Data preparation	Keeps fields analysis-ready and documents cleaning decisions.	Tableau Data Source / spreadsheet
Analytics layer	Aggregates measures for comparisons and KPIs.	Tableau: SUM(sales), AVG(range), AVG(rating), counts
Visualization worksheets	Answers focused business questions using individual charts.	Bar, pie/treemap, scatter, geographic and year/category charts
Dashboard interface	Combines KPI cards and charts with filters and tooltips.	Tableau dashboard, filters, legends
End user	Interprets market patterns and compares segments.	Student, reviewer, or business analyst
Publishing and access	Shares the dashboard for review when required.	Tableau Public; published workbooks are public